Engine Model(s): PT6A

72-00-00

ENGINE, TURBOPROP - SERVICING

1. General

CAUTION: TO PREVENT POSSIBLE BEARING DAMAGE SUCH AS BRINELLING, USE OF A "PRATT AND WHITNEY CANADA APPROVED" SHIPPING/STORAGE CONTAINER/SKID WITH SHOCK MOUNTS IS NECESSARY FOR ALL TRANSPORTATION OF AN ENGINE (REF. IPC P/N 3013244, 72-00-00).

A. The following information and procedures are necessary for storage, preservation and depreservation of an engine or power section, and removal and installation in an appropriate fiberboard/reinforced fiberboard container, metal container or stand.

NOTE: For engine installation in airframe, refer to REMOVAL/INSTALLATION.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC03-001	Oil, Engine Lubricating
PWC05-053	Polyethylene Sheet
PWC05-054	Bag, Polyethylene
PWC05-070	Tape, Pressure Sensitive
PWC05-077	Oil, Preservative
PWC05-094	Oil, Gas Turbine, Synthetic Lubricating and Preservative
PWC05-094A	Oil, Gas Turbine, Synthetic Lubricating and Preservative
PWC05-107	Foam, Polyurethane
PWC06-007	Lubricant, Dry Film
PWC06-012	Compound, Preservative
PWC09-003	Compound, Sealing

3. Special Tools

The special tools listed below are used in the following procedures.

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Tool No.	Name	Application
PWC30077	Puller	
PWC30128-4	Puller	
PWC30269	Mount Ring	
PWC32080	Pad, Engine Mount	
PWC32081	Ring, Mounting	Replaced by PWC70334
PWC34300	Stand, Engine	
PWC34752	Adapter, Engine Mount	Replaces PWC30712
PWC50373	Bracket, Lifting	
PWC51140	Stand Assembly, Engine	Obsolete - Replaced by PWC70907
PWC51861-600	Sling, Engine	
PWC70334	Ring, Mounting	Replaces PWC32081
PWC70792	Adapter	
PWC70907	Stand Assembly, Engine	Obsolete - Replaced by PWC71835
PWC71835	Build Stand, Engine Assembly	Alternate to PWC70907

4. Fixtures, Equipment and Supplier Tools

Not Applicable

5. Lubricating Oil System

A. Lubricating Oil

The lubricating oils specified for use in PT6A turboprop engines are detailed in Pratt & Whitney Canada, (P&WC) Service Bulletin 1001 and will be revised periodically to include recently approved oils. All oils listed in the bulletin are for use in commercially operated engines, and are approved for flight operation.

In cases when oils approved by P&WC are not available and other oils have to be substituted, an operator must obtain prior approval or recommendations for use of such oil from:

Pratt & Whitney Canada Corp.
Att. Customer Support Services
1000 Marie-Victorin
Longueuil, Quebec
Canada
J4G 1A1

B. Oil Level Check

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To avoid overfilling of oil tank, and high oil consumption, an oil level check is recommended within 30 minutes after engine shutdown. Ideal interval is 15 to 20 minutes. If more than 30 minutes has passed, and the dipstick indicates that oil is needed, start the engine and run at ground-idle for five minutes, and recheck oil level as follows:

C. Procedure

(1) Unlock filler cap and dipstick from filler neck at 11 o'clock position on accessory gearbox and remove filler cap.

(2) Wipe the dipstick with clean lint free cloth.

CAUTION: WHEN THE FILLER CAP AND DIPSTICK/GAGE ASSEMBLY IS INSTALLED AND LOCKED, NO MOVEMENT IS ALLOWED.

(3) Install the cap/dipstick and lock.

(4) Remove the cap/dipstick.

(5) Check the oil tank contents against the markings on dipstick (markings correspond to U.S. quarts) and service as required.

NOTE: 1. Normal oil level is one U.S. quart (0.83 Imp. quart, 0.95 liter) below maximum level with engine in horizontal altitude.

NOTE: 2. Filling the oil to the maximum level may result in high consumption rate, with the oil exiting through the AGB breather. On some engines, this may also occur with the oil level at one or two US quarts below the maximum level. In such cases, operators are advised to service the oil to the level that results in acceptable consumption, down to 3 quarts below the maximum, if necessary. For engines with an oil sight gauge, keep the level within the green band. This practice is acceptable, due to the large usable oil quantity, and providing the oil level is monitored using the engine maintenance manual, making sure the consumption allowance and operation are within the recommended oil temperature and pressure.

(6) If oil level is too low to register on dipstick due to possible excessive consumption, or if low or fluctuating pressures have been recorded, refer to Fault Isolation, Engine Lubrication, for action to be taken, then proceed as follows:

(a) Fill oil tank to normal level and record quantity of oil added.

NOTE: If oil overflows during replenishing, verify the sight gage for oil level. For engines without an oil sight gage and with an oil filler ball valve, inspect ball valve for freedom of movement. Replace the oil filler and ball valve if required (Ref. 72-60-00, ACCESSORY GEARBOX ASSEMBLY - MAINTENANCE PRACTICES).

CAUTION: WHEN FILLER CAP ASSEMBLY IS INSTALLED AND LOCKED, NO MOVEMENT IS ALLOWED.

(b) Install filler cap/dipstick, make sure cap is locked.

(c) Run engine at ground-idle for approximately five minutes (Ref. 71-00-00,

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ADJUSTMENT/TEST).

(d) Check oil level (Ref. Steps (1) through (6) preceding).

D. Oil Servicing (Ref. Fig. 301)

CAUTION: DO NOT MIX DIFFERENT VISCOSITIES OR SPECIFICATIONS OF OIL AS THEIR DIFFERENT CHEMICAL STRUCTURE CAN MAKE THEM INCOMPATIBLE.

NOTE: For oil change recommendations, refer to SB1001.

(1) Oil System Draining

(a) Place large drip pan below engine and suitable oil container under each drain point.

(b) Remove lead from chip detector (3) (Ref. Aircraft Maintenance Manual).

(c) Remove drain plug (1) (Pre-SB1217) or chip detector (3) (Post-SB1217) from propeller reduction gearbox and discard preformed packings (Ref. 72-10-00).

(d) Remove drain plug from accessory gearbox and discard preformed packing (Ref. 72-60-00).

(e) Remove the cotterpin (9) and flat head pin (6) from compressor inlet case. Remove the drain plug (8) with puller (PWC30128-4) or (PWC30077) and discard the preformed packing (7).

NOTE: Refer to the Aircraft Maintenance Manual for procedure on draining the airframe side of oil system.

(2) Oil System Filling

(a) Install preformed packing on drain plug (Pre-SB1217) or chip detector (Post-SB1217) and screw into propeller reduction gearbox. Torque plug or chip detector 45 to 55 lb.in. and safety wire (Ref. 72-10-00).

(b) Install preformed packing on drain plug and screw into accessory gearbox. Torque plug 200 to 225 lb.in. and safety wire (Ref. 72-60-00).

(c) Install preformed packing on oil tank drain plug. Press drain plug into inlet case. Fasten with flat head pin and lock with cotterpin.

(d) Refill oil tank with specified oil (Ref. SB1001).

NOTE: Refer to Subparagraph B. for checking oil tank levels.

(e) Install filler cap and dipstick (Ref. 72-60-00, Removal/ Installation).

(f) Replace aircraft lead on chip detector, torque lead connector and lockwire (Ref. Aircraft Maintenance Manual).

E. Oil System Flushing

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CAUTION: IF THE ENGINE OIL SYSTEM HAS BEEN FILLED WITH AN UNAPPROVED OIL TYPE OR IF THE OIL SYSTEM HAS BEEN CONTAMINATED BY OTHER THAN METALLIC MATTER, THE SYSTEM MUST BE FLUSHED AND FILLED AGAIN WITH THE ORIGINAL OIL TYPE OR AN APPROVED OIL TYPE.

(1) Place suitable containers or drip pan under engine.

CAUTION: LIMIT ENGINE ROTATION TO THE MINIMUM TIME REQUIRED TO DO COMPLETE DRAINING. ALSO, OBSERVE STARTER OPERATING LIMITATIONS.

(2) With drains open, place the starting control lever to CUT-OFF and the IGNITION switch to OFF. Motor the engine with the starter only and allow the scavenge pumps to clear all lubricating oil.

(3) Reinstall all the drain plugs and chip detector (Ref. Subpara. D.).

(4) Refill the engine oil tank (Ref. Subpara. C.).

(5) Start the engine and run at idle speed (Ref. 71-00-00, ADJUSTMENT/TEST) for a minimum of two minutes.

(6) Feather the propeller.

(7) Shut down the engine (Ref. 71-00-00, ADJUSTMENT/TEST).

(8) Repeat steps (1) through (2) , preceding.

(9) Remove the main oil filter to clean or replace with a new filter (Ref. 79-20-02, OIL FILTER, HOUSING AND CHECK VALVE - MAINTENANCE PRACTICES). Then install the oil filter again.

(10) Remove the reduction gearbox scavenge oil strainer and clean (Ref. 72-10-00). Reinstall the strainer.

(11) Reinstall all engine drain plugs and chip detector. Tighten and lockwire (Ref. Subpara. D.).

(12) Repeat steps (4) through (7) , preceding.

(13) Check the oil level and replenish, as necessary (Ref. Subpara. C.).

CAUTION: WHEN THE FILLER CAP AND DIPSTICK/GAGE ASSEMBLY IS INSTALLED AND LOCKED INTO POSITION, NO MOVEMENT IS ALLOWED.

(14) Install the filler cap and dipstick/gage assembly in filler tube. Make sure the cap is correctly installed and locked securely.

F. Procedure After Oil System Flushing

CAUTION: DIFFERENT FORMULATIONS OF VARIOUS OIL BRANDS MAY HAVE DIFFERENT DETERGENT ACTION. AFTER AN OIL BRAND CHANGE, THIS MAY CAUSE RELEASE OF CARBON PARTICLES INTO THE OIL SYSTEM RESULTING IN BLOCKAGE OF THE SCAVENGE SCREEN.

(1) After a change of oil brand, the main oil filter should be inspected for carbon particles at 10 hour intervals up to a total of 50 hours (5 inspections) and at routine oil filter checks thereafter, up to 500 hours.

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(2) When carbon in excess of normal amount is noted:

(a) Place a suitable drip pan under the accessory gearbox or engine.

(b) Remove the oil drain plug from the 6 o'clock position on the accessory gearbox rear housing.

(c) Inspect the scavenge pump screen through the drain hole using a suitable mirror and light.

(d) If carbon is present, attempt to remove using a small stiff brush.

(e) Flush out any particles of removed carbon.

(f) If carbon cannot be removed by this method, remove the accessory gearbox (Ref. 72-60-00) and clean the screen.

G. Oil Level Check and Replenishment following Multiple Motoring Runs or Wind-Milling of a Shutdown Engine in Flight

(1) At low gas generator speeds (Ng) during motoring runs or wind-milling of a shutdown engine in flight, oil leakage can occur from the compressor inlet case and in some cases the gas generator drains and exhaust duct. This is a result of the main oil pump efficiency being higher than the scavenge pump efficiency at low gas generator speeds (Ng), causing the accessory gearbox (AGB) and bearing cavities to flood with oil.

(a) Procedure:

1 Take note of the oil level. Do not add oil.

2 Place suitable container under engine.

3 Remove accessory gearbox oil drain plug and allow oil to drain completely.

4 Install drain plug.

5 Clean compressor inlet case and any other areas wet with oil (Ref. 71-00-00, POWER PLANT - CLEANING).

6 Check oil level and fill as necessary.

7 Start engine and run at FLIGHT - IDLE (Ref. 71-00-00, ADJUSTMENT/TEST, Engine Starting) for a minimum of 10 minutes.

8 Check for oil leaks.

NOTE: It is normal to have additional oil leakage as the oil is scavenged back to the oil tank.

9 Repeat steps 5 thru 8 above. If oil leak stops, return engine to service, otherwise refer to fault isolation for additional troubleshooting instructions.

H. P&WC's Oil Analysis Technology Sampling Procedure

(1) General

(a) This procedure is to provide instructions for collection of oil samples for the P&WC's oil

analysis technology program.

NOTE: 1. P&WC's oil analysis technology is an optional task. It can be done at the discretion of the operator.

NOTE: 2. The oil sampling kits are available through P&WC customer portal. Refer to P&WC SIL GEN-153 or contact P&WC for more information.

NOTE: 3. P&WC's oil analysis technology is complementary to oil filter/chip analysis, it is not replacing it. You can do the oil filter debris and chip analysis in any independent laboratory approved by P&WC per SIL GEN-123. Do chip/filter analysis during scheduled maintenance (filter and chip inspection) or in an unscheduled maintenance activity.

(b) Oil sampling is recommended within 15 to 30 minutes after engine shutdown. If the time after engine shutdown is more than 30 minutes, start the engine and operate at ground idle for 5 minutes or until oil temperature is stable.

(c) Collect the oil sample before you do the engine oil service.

(d) After an engine oil system flush, a minimum of 20 hours of engine operation is recommended before you take an oil sample.

(2) Recommended Interval

(a) For operators enrolled in the P&WC's oil analysis technology program, it is recommended to take oil samples at aircraft MINOR interval, or annually, whichever occurs first.

NOTE: In order to be valid, oil sampling must be done in accordance with the engine oil system oil analysis procedure (Ref. Step (3)).

(b) Based on oil analysis results, P&WC can recommend a different sampling interval for a specific engine.

(c) In case of an unplanned removal for an oil system related issue, take an oil sample before removal if possible.

(3) Oil Sampling Procedure:

CAUTION: MAKE SURE THE AREA IS CLEAN IN ORDER TO PREVENT CONTAMINATION OF THE ENGINE OIL SYSTEM AND OIL SAMPLE. MAKE SURE THAT THE CONTAINER AND TOOLING ARE FREE OF DEBRIS AND CONTAMINANTS.

(a) Wait 15 to 30 minutes after engine shutdown (Ref. 71-00-00, POWER PLANT - ADJUSTMENT/TEST).

(b) Make sure that the plastic bottle is clean and dry.

NOTE: You can clean the plastic bottle with water and dry with compressed air.

(c) Record the applicable information on the bottle label and attach it on the bottle.

(d) Unlock the filler cap and dipstick from the filler neck at the 11 o'clock position on the

accessory gearbox. Remove the filler cap.

(e) Collect a sample of 60 ml (2 oz.) engine oil through the filler neck with the syringe and the flexible tube.

(f) Put the oil sample collected with the syringe into the plastic bottle.

NOTE: Do not use syringe and flexible tubes more than one time. Discard the used syringe.

(g) Make sure that the plastic bottle is sealed correctly and put it in a separate plastic bag supplied with the oil sampling kit.

(h) Complete the oil analysis sampling form P&WC 11914 or 12233 supplied in the oil sampling kit.

NOTE: Use one form for each engine sample.

(i) Put the sealed plastic bag with the oil sample bottle and the completed oil analysis sampling form into the shipping envelope and send in accordance with P&WC SIL GEN-153.

(j) Do a check for the engine oil level and fill as necessary (Ref. Subpara. C.).

6. Preservation, Depreservation and Unpreserved Engines Section

A. General

Preservation of engines in service depends on the period of inactivity and whether or not the engine may be rotated during the inactive period. An engine is considered inactive when it has not been operated either on the ground, or in flight for a minimum of ten minutes after the oil temperature has stabilized. The expected period of inactivity should be established and reference made to the Engine Preservation Procedures following (Ref. Para. B.).

The preservation done should be recorded in the engine log book and on tags fastened to the engine.

For an engine inactive in a severe environment such as very high temperature changes, high humidity, salt-laden or harsh atmosphere, it is recommended to preserve the engine to the next higher schedule or start the engine and operate more frequently for 10 minutes minimum.

If an engine has been preserved for a certain schedule (91 days to 1 year), and now the engine will be inactive for a longer period of time (more than 1 year), the engine should be depreserved, inspected and tested according to the 91 days to 1 year depreserving instructions. Then, remove the engine, preserve according to schedule and store in an engine container.

B. Engine Preservation Procedure

CAUTION: ENGINE(S) REMOVED FROM SERVICE AND NOT PRESERVED PER THE PUBLISHED PROCEDURES, COULD POSSIBLY SUFFER CORROSION DAMAGE. THE ONLY METHOD TO DETERMINE IF SUCH DAMAGE HAS OCCURRED, IS TO DISASSEMBLE THE ENGINE PARTIALLY FOR VISUAL INSPECTION OF ALL COMPONENTS.

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CAUTION: DO NOT SPRAY PRESERVATIVE OIL INTO COMPRESSOR OR EXHAUST PORTS OF ENGINES. DIRT PARTICLES DEPOSITED ON BLADES AND VANES COULD ADHERE AND ALTER AIRFOIL SHAPE, ADVERSELY AFFECTIVE COMPRESSOR EFFICIENCY.

(1) Engines inactive 0 to 7 days.

(a) If the engine operated in a salt-laden or harsh environment, do a compressor and turbine desalination wash (Ref. 71-00-00, POWER PLANT - CLEANING).

(b) Engine may remain inactive with no preservation protection provided engine is sheltered, humidity is not excessively high and there is not extreme temperature changes that may produce condensation.

(c) Install inlet and exhaust covers.

(d) When propeller is removed, apply a coating of preservative oil (PWC05-094) or (PWC05-094A) to the inner bore of the propeller shaft and seal off the opening with a clean plastic film.

(e) When an engine mounted accessory is removed, apply a coating of preservative oil (PWC05-094) or (PWC05-094A) on the exposed drive pad and gearshaft, then install shipping cover.

(2) Engines inactive 8 to 28 days.

(a) Do 0 to 7 days procedures.

(b) Place desiccant bags and humidity indicator on wooden racks in engine exhaust duct only.

(c) Seal off all engine openings. Make sure exhaust cover has suitable window to monitor humidity indicators.

(d) Do a check for relative humidity every two weeks if the engine is stored outside, and every 28 days if the engine is stored inside. Relative humidity must be maintained at 40%. If the humidity indicator turns pink, replace desiccant bags and the indicator, find the reason for humidity and remove the source of humidity.

(3) Engines inactive 29 to 90 days.

(a) Do 0 to 7 days, and 8 to 28 days procedures.

(b) Wash engine externally.

(c) Do a compressor performance recovery/desalination wash and a turbine rinse (Ref. 71-00-00, CLEANING).

(d) Examine all engine external protective coating, and touch-up/repair as necessary.

(e) Lubricate all linkages.

(f) Place a suitable container under the engine.

(g) Disconnect fuel inlet to oil-to-fuel heater and connect suitable oil supply line to oil-to-fuel heater fuel inlet. Blank off disconnected fuel supply line.

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(h) Disconnect fuel line at flow divider and dump valve (PT6A-21) or from inlet of manifold adapter (PT6A-27 and PT6A-28) to prevent preservation oil from entering fuel manifold. Blank off elbow on adapter or valve, as applicable.

CAUTION: UNDER NO CIRCUMSTANCES PERMIT PRESERVATIVE OIL TO ENTER ENGINE WHERE IT MAY COME INTO CONTACT WITH THERMOCOUPLE PROBE ASSEMBLY. OIL CONTAMINATION OF PROBES MAY CAUSE COMPLETE FAILURE OF THERMOCOUPLE SYSTEM.

(i) Supply preserving oil (PWC05-077) at 5 to 25 psig at least 16°C (60°F) to fuel supply line on oil-to-fuel heater.

CAUTION: OBSERVE STARTER MOTOR OPERATING LIMITS (REF. STARTER MANUFACTURER'S MANUAL).

(j) With ignition system OFF, fuel condition lever to GROUND-IDLE and power control lever to TAKE-OFF, carry out normal motoring run until all preservative oil is displaced. During motoring run, move power control lever from TAKE-OFF to GROUND-IDLE and back to TAKE-OFF and fuel condition lever from GROUND-IDLE to OFF and back to GROUND-IDLE to displace fuel from system.

(k) After motoring run, check to see if preservative oil is coming from opened fuel line. If not, repeat motoring cycle until preservative oil flows from opened fuel line.

(l) Return power control lever to GROUND-IDLE and fuel condition lever to OFF. Reconnect fuel supply to oil-to-fuel heater and fuel line to flow divider inlet.

(m) Install caps, covers and plugs as necessary to prevent entry of foreign material and accumulation of moisture .

(4) Engines inactive for periods exceeding 90 days.

(a) Do 0 to 7, 8 to 28 and 29 to 90 days procedures.

CAUTION: OBSERVE STARTER MOTOR OPERATING LIMITS (REF. STARTER MANUFACTURER'S MANUAL).

(b) Close normal fuel supply shutoff valve and motor engine with starter until oil pressure and Ng are indicated. Disengage starter.

(c) Place suitable container under the engine.

(d) Drain engine oil.

(e) With drains open, motor engine (Ref. 71-00-00, ADJUSTMENT/TEST) to permit scavenge pumps to clear engine, indicated by cessation of steady stream of oil from drains. To prevent excessive operation with limited lubrication, limit rotation to shortest possible time to do complete draining.

(f) Remove oil filter element (Ref. 79-20-02) and allow to drain.

(g) Let oil to drain to a slow drip, approximately one drip per half hour, then install oil filter and close drains.

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(h) Remove covers from accessory drive pads. Spray exposed surfaces and gearshafts with the preservative oil (PWC05-094) or (PWC05-094A). Install the covers again on the accessory drive pads.

CAUTION: COMPOUND MUST NOT TO BE APPLIED TO ANY BOLT THREADS USED TO RETAIN ACCESSORIES. THESE THREADS MUST BE COATED WITH REGULAR ENGINE OIL AS REQUIRED BY NORMAL TIGHTENING PROCEDURES. UNDER NO CIRCUMSTANCES MUST THE COMPOUND BE USED IN, OR PERMITTED TO ENTER ANY AREA WHERE IT COULD CONTAMINATE THE ENGINE LUBRICATION SYSTEM.

(i) Coat all external flanges, bolts, and studs with compound (PWC09-003). In particular, pay special attention to the RGB and compressor inlet case flanges.

(j) Install caps, covers and plugs, as necessary to prevent entry of foreign material and accumulation of moisture .

(k) Tag oil filler cap with date of preservation and enter date and type of preservation in engine log book.

(l) Install humidity indicator in air inlet and exhaust of engine compartment. Cover with suitable airtight moisture barrier. Provide inspection windows at each end for observation of humidity indicators.

NOTE: Inspection of a preserved unit should be carried out every two weeks if aircraft is stored outside, or every 30 days if aircraft is stored inside. If relative humidity, as indicated on humidity indicator, is less than 40 percent, no further action is required. If humidity indicated exceeds 40 percent, desiccant bags must be replaced or reactivated (Ref. Para. 7. D.).

(5) Engines inactive for periods exceeding one year:

(a) Do 0 to 7, 8 to 28, 29 to 90, and 91 days and over procedures.

(b) Remove the engine and store it in a shipping/storage container.

(6) Fuel System Preservation

(a) Fuel supply and ignition switches OFF.

(b) Place a suitable container under the engine.

(c) Disconnect fuel inlet line at oil-to-fuel heater and blank off line.

CAUTION: THE FOLLOWING EQUIPMENT MUST BE SUPPLIED WITH A SUITABLE FILTER NO COARSER THAN 10-MICRON RATING, TO PREVENT FOREIGN MATERIAL FROM BEING DRAWN INTO ENGINE FUEL SYSTEM.

(d) Connect suitable oil supply line to oil-to-fuel heater fuel inlet.

(e) Disconnect fuel line from inlet manifold adapter (PT6A-27 and PT6A-28) (Ref. 73-10-04) or flow divider and dump valve (PT6A-21). Blank off elbow on adapter or valve, as applicable.

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CAUTION: DO NOT ALLOW PRESERVATIVE OIL TO ENTER ENGINE WHERE IT MAY COME IN CONTACT WITH THERMOCOUPLE PROBES. OIL CONTAMINATION OF PROBES MAY CAUSE COMPLETE FAILURE OF THERMOCOUPLE SYSTEM.

(f) Connect a supply of preservative oil (PWC05-077) at 5 to 25 psig, 16 °C (60 °F) to fuel supply line on oil-to-fuel heater (Ref. Step (d) preceding).

CAUTION: TO PREVENT EXCESSIVE OPERATION WITH LIMITED LUBRICATION, MAINTAIN MOTORING TO THE SHORTEST POSSIBLE TIME. OBSERVE STARTER MOTOR OPERATING LIMITS (REF. TO STARTER MANUFACTURER'S MANUAL).

(g) Motor engine with starter and advance Condition Lever (PT6A-21) or Starting Flow Control Lever (PT6A-27 and PT6A-28) to OPEN and Power Lever to IDLE.

(h) After motoring run, return levers to the OFF or CLOSED position.

(i) If no oil is flowing from open fuel line, refer to starter manufacturer's manual for time interval before repeating procedure.

(j) Remove blanking plug from fuel line, and oil supply line from oil-to-fuel heater fuel inlet. Connect fuel supply line to oil-to-fuel heater. Torque coupling nut (Ref. Airframe Manufacturer's Maintenance Manual) and lockwire.

(k) Remove blanking cap from flow divider and dump valve elbow (PT6A-21) or from inlet manifold adapter (PT6A-27 and PT6A-28) and connect fuel line. Torque coupling nut 90 to 100 lb.in. and lockwire.

(l) Complete preservation.

(7) Oil System Preservation

(a) Make sure fuel supply and ignition are OFF.

CAUTION: TO PREVENT EXCESSIVE OPERATION WITH LIMITED LUBRICATION, MAINTAIN MOTORING TO SHORTEST POSSIBLE TIME. OBSERVE STARTER MOTOR OPERATING LIMITS (REF. STARTER MANUFACTURER'S MANUAL).

(b) Motor engine until oil pressure and compressor speed (Ng) is indicated; stop motoring.

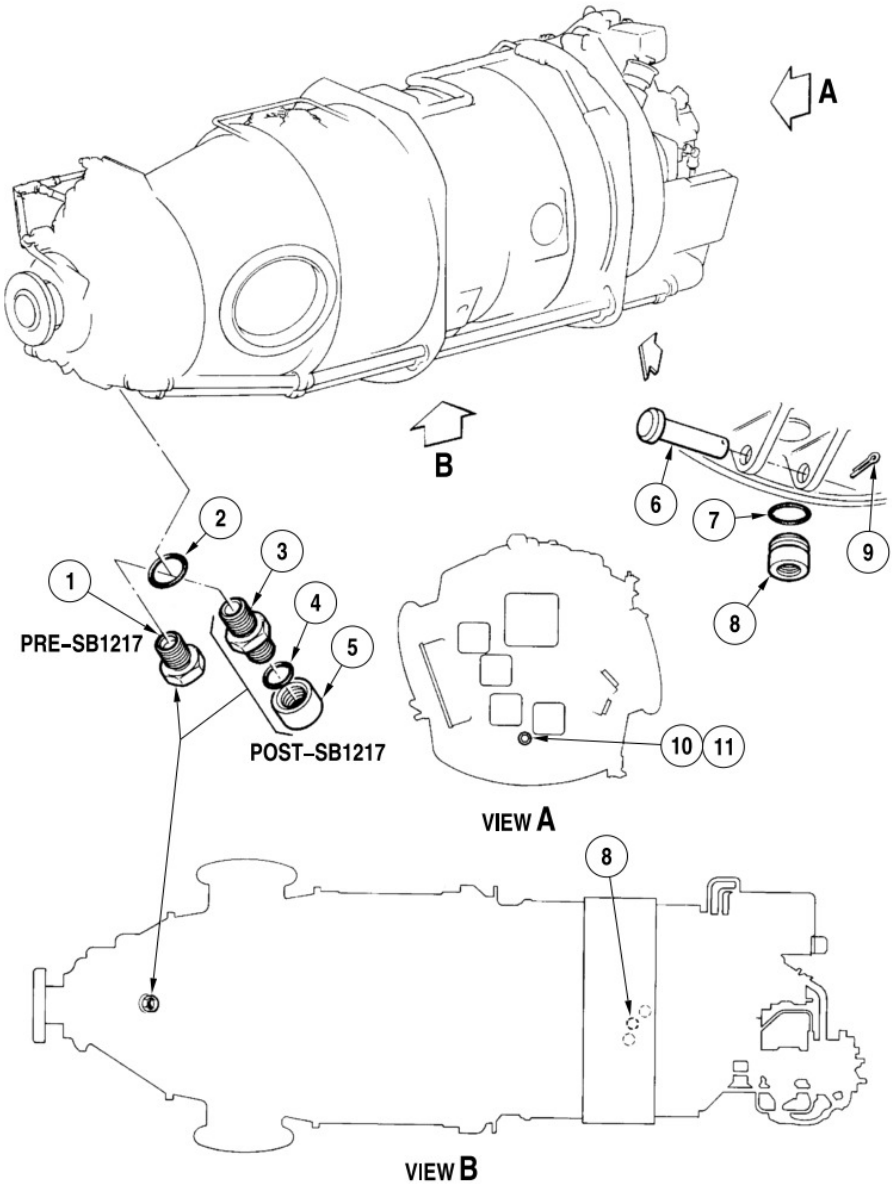
(c) Place large drip pan below engine and suitable oil container under each drain point.

(d) Remove lead from chip detector (Ref. Aircraft Maintenance Manual).

(e) Remove drain plug (1, Fig. 301, Pre-SB1217) or chip detector (3, Post-SB1217) from propeller reduction gearbox and discard preformed packings (2) and (4).

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Figure 301 Location of Oil Drain Plugs



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1. Propeller Reduction Gearbox Drain Plug (Pre-SB1217)
2. Preformed Packing
3. Chip Detector (Post-SB1217)
4. Preformed Packing (used for storage and shipping)
5. Cover (Post-SB1217) (used for storage and shipping)
6. Flat Head Pin
7. Preformed Packing
8. Oil Tank Drain Plug
9. Cotterpin
10. Accessory Gearbox Drain Plug
11. Preformed Packing

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(f) Remove drain plug (10) from accessory gearbox and discard preformed packing (11).

(g) Remove cotterpin (9) and flat head pin (6) from compressor inlet case extract drain plug (8) from inlet case with sling (PWC30037) and discard preformed packing (7).

(h) Motor engine to allow scavenge pumps to clear oil, this will be indicated by slowing down, then stopping of the steady stream of oil from each drain point; stop motoring.

(i) Remove oil tank filter element from inlet case (Ref. 79-20-04,MAINTENANCE PRACTICES).

(j) Allow oil to drain for one-half hour, then reinstall filter (Ref. 79-20-04, MAINTENANCE PRACTICES).

(k) Install new preformed packing (2) on drain plug (1, Fig. 301, Pre-SB1217) or new preformed packings (2 and 4) on chip detector (3, Post-SB1217) and install in propeller reduction gearbox.

(l) Install new preformed packing (11) on drain plug (10) and install in accessory gearbox.

(m) Install new preformed packing (7) on drain plug (8) and install in compressor inlet case. Install flat head pin (6) and cotterpin (9).

C. Engine Depreservation Procedure

CAUTION: UNDER NO CIRCUMSTANCES PERMIT PRESERVATIVE OIL TO ENTER ENGINE WHERE IT MAY COME INTO CONTACT WITH THERMOCOUPLE PROBE ASSEMBLY. OIL CONTAMINATION OF PROBES MAY CAUSE COMPLETE FAILURE OF INDICATING SYSTEM.

(1) 0 to 7 days:

(a) No depreservation required. Remove covers from inlet and exhaust and check for obstructions.

(2) 8 to 28 days:

(a) Remove desiccant bags, humidity indicator and moisture barrier.

(b) Make sure previously sealed engine openings are reopened and are unobstructed.

(3) 29 to 90 days:

(a) Remove engine intake and exhaust covers, moisture barriers, desiccant bags and humidity indicators. Depreserve engine fuel system.

(4) 91 days to one year:

(a) Remove all moisture barriers, desiccant bags and humidity indicators.

(b) Open previously sealed openings.

(c) Check for any obstructions

(d) Inspect all external engine cases, flanges, accessories and hardware for evidence of corrosion.

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(e) Remove the compressor inlet screen and inspect the inlet case and visible flanges internally for evidence of corrosion. If corrosion is found, inspect the accessory gearbox and the reduction gearbox internally for corrosion with a borescope.

(f) If corrosion is outside limits, remove the engine and overhaul for inspection after storage. Slowly rotate the propeller by hand and check for any stiffness, audible scraping, binding or rubbing.

(g) Service the engine oil system (Ref. Para. 5.).

(h) Depreserve the fuel system (Ref. Para. 6.D.).

NOTE: During motoring, check for stiffness, audible scraping, binding or rubbing.

(i) After the engine run, inspect all gas generator and exhaust case drain valves for presence of oil. If oil is evident, overhaul the engine for inspection after storage.

(j) Inspect all fuel filters. Clean or replace as necessary.

(k) Inspect the oil filter, reduction gearbox strainer and chip detector(s) for contamination. If contamination is evident, and the engine has no previous recent contamination history, remove and overhaul the engine. Otherwise refer to Chapter 79-20-04.

(5) One year and over:

NOTE: The storage life limit of the Honeywell Fuel Control Unit (FCU) is six years. If the storage of the FCU is more than six years, send the FCU for overhaul.

(a) Engine must be completely depreserved and lubrication system serviced.

D. Fuel System Depreservation

(1) Overhauled engines and fuel control units are subjected to preservation procedure to protect them during shipping and storage. When the engine is installed in the aircraft, or the fuel control unit installed on the engine, the fuel system is depreserved:

(a) Disconnect fuel lines from inlet manifold adapter (PT6A-27 and PT6A-28) or flow divider and dump valve (PT6A-21). Blank off elbow on adapter or valve, as applicable.

(b) Place FCU Condition Lever (PT6A-21) or Starting Flow Control Lever (PT6A-27 and PT6A-28 engines) to OFF or CLOSED.

(c) Fuel ON.

(d) Boost pump ON.

(e) Fuel pressure 5 psig minimum.

(f) Power Control Lever CLOSED.

(g) Ignition OFF.

CAUTION: MAINTAIN MOTORING PERIOD TO SHORTEST POSSIBLE TIME. OBSERVE STARTER MOTOR OPERATING LIMITS (REFER TO AIRCRAFT OR STARTER MANUFACTURER'S MANUAL).

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CAUTION: DO NOT ALLOW PRESERVATIVE OIL TO CONTACT THERMOCOUPLE PROBES. OIL CONTAMINATION OF PROBES MAY CAUSE COMPLETE FAILURE OF INDICATING SYSTEM.

(h) Motor engine with starter and advance Condition Lever or Starting Flow Control Lever to OPEN position and Power Lever to IDLE.

(i) When clean fuel, free of oil flows from open line, return power lever to OFF or CLOSED, and stop motoring.

(j) Remove blanking cap from elbow and connect fuel lines. Torque line coupling nut 90 to 100 lb.in. and lockwire.

(k) Fuel OFF, boost pump OFF.

E. Engine Unpreserved Procedure

NOTE: 1. Do the steps that follow for inactive engines not preserved as recommended in the "Engine Preservation Procedure".

NOTE: 2. If compressor or turbine wash performed, do a dry motoring and subsequent engine run (Ref. 71-00-00, POWER PLANT - ADJUSTMENT/TEST).

(1) 0 to 7 days inactivity:

(a) If the engine operated in a salt-laden or harsh environment, do a compressor desalination wash and a turbine wash (Ref. 71-00-00, POWER PLANT - CLEANING).

(b) Make sure all engine openings are clear and unobstructed and no evidence of corrosion. Inspect engine externals for condition and corrosion.

(c) If further storage is required, preserve depending on the expected period of inactivity per applicable EMM.

(2) 8 to 28 days inactivity:

(a) Do 0 to 7 days unpreserved engine procedure requirement.

(b) For engines stored in an air conditioned environment, go to Step (d).

NOTE: An air conditioned unit is defined as an enclosed climate controlled environment with temperature maintained above 8 °C (46 °F) and not susceptible to rapid temperature changes.

(c) For other engines, perform the following steps:

1 Prior engine start, do a Total Acid Number (TAN) test of oil in engine per related lubricating oil **SB1001**.

2 If TAN test is found acceptable, proceed to engine run (Ref. Step (d)).

3 If TAN test is above the limits, inspect the internal of the AGB and RGB using a borescope for presence of corrosion or internal coating flaking. If corrosion and/or flaking are found, remove the engine and send it to an approved facility for light overhaul.

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4 If TAN test is above limits, drain and flush oil (Ref. Para. 5. E.).

(d) Run the engine at 80% TO power for 10 minutes.

(e) Remove and inspect oil filter and chip detector(s). If anything abnormal is found, refer to applicable EMM.

(f) Do a check for evidence of fuel and oil leaks at propeller shaft seal, accessory mounting pads, external tubes and hoses.

(g) If further storage is required, preserve depending on the expected period of inactivity per applicable EMM.

(h) After return to service, repeat inspection of oil filter and chip detector(s) after 10 hours. If anything abnormal is found, refer to applicable EMM otherwise return to standard periodic interval.

(3) 29 to 90 days inactivity:

(a) Do 0 to 7 days unpreserved engine procedure requirement.

(b) Prior engine start, do a Total Acid Number (TAN) test of oil in engine per related lubricating oil **SB1001**.

1 If TAN test is found acceptable, proceed to drain and flush oil (Ref. Step (c)).

2 If TAN test is above the limits, inspect the internal of the AGB and RGB using a borescope for presence of corrosion or internal coating flaking. If corrosion and/or flaking are found, remove the engine and send it to an approved facility for light overhaul.

3 If TAN is above the limits, replace the oil filter.

(c) Drain and flush oil (Ref. Para. 5. E.).

(d) Rotate propeller with your hand and check for binding. If binding, inspect per applicable EMM requirements (Ref. Inspection).

(e) Inspect the engine externals and linkages for condition and evidence of corrosion.

(f) Inspect the compressor for corrosion and cleanliness. Do compressor desalination wash, if dirty or salt laden (Ref. 71-00-00, POWER PLANT - CLEANING).

(g) Inspect the fuel filter.

(h) Disconnect fuel tube at inlet of flow divider and flush fuel system per a wet motoring run.

(i) Run engine at 80% TO power for 10 minutes and do acceleration check.

(j) Remove and inspect internal AGB scavenge pump inlet screen, oil filter and chip detector(s). If anything abnormal is found, refer to applicable EMM.

(k) Check for evidence of fuel and oil leaks at propeller shaft seal, accessory mounting pads, external tubes and hoses.

(l) If further storage is required, preserve depending on the expected period of inactivity

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per applicable EMM.

(m) After return to service, repeat the inspection of oil filter and chip detector(s) after 10 and 50 hours. If anything abnormal is found, refer to applicable EMM otherwise return to standard periodic interval.

(4) For period greater than 91 days to less than 1 year:

(a) Do the 0 to 7 day unpreserved engine procedure requirement.

(b) Prior engine start, do a Total Acid Number (TAN) test of oil in engine per related lubricating oil **SB1001**.

1 If TAN test is above limits, remove the engine and send to an approved facility for light overhaul.

2 If TAN is acceptable, inspect the internal of the AGB and RGB using a borescope for presence of corrosion or internal coating flaking. If corrosion and/or flaking are found, remove the engine and send to an approved facility for light overhaul.

(c) Replace the oil filter.

(d) Drain and flush oil (Ref. Para. 5. E.).

(e) Rotate propeller with your hand and check for binding. If binding, inspect per applicable EMM requirements (Ref. Inspection).

(f) Inspect the engine externals and linkages for condition and evidence of corrosion.

(g) Inspect the compressor for corrosion and cleanliness. Do compressor desalination wash, if dirty or salt laden (Ref. 71-00-00, POWER PLANT - CLEANING).

(h) Inspect fuel filter.

(i) Clean fuel nozzles, perform leak and functional test (Ref. 73-10-05, FUEL MANIFOLD AND NOZZLES - MAINTENANCE PRACTICES).

(j) Route FCU to an approved shop for inspection and functional test.

(k) Once the FCU installed (original unit or replacement), disconnect fuel tube at inlet of flow divider and flush fuel system per a wet motoring run.

(l) Run engine at 80% TO power for 10 minutes and do acceleration check.

(m) Remove and inspect internal AGB scavenge pump inlet screen, oil filter, and chip detector(s). If anything abnormal is found refer to applicable EMM.

(n) Check for evidence of fuel and oil leaks at propeller shaft seal, accessory mounting pads, external tubes and hoses.

(o) If further storage is required, preserve depending on the expected period of inactivity per applicable EMM.

(p) After return to service, repeat inspection of oil filter and chip detector(s) after 10 and 50 hours. If anything abnormal is found, refer to applicable EMM otherwise return to standard periodic interval.

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(5) For periods of 1 year and greater (regardless of humidity and environment):

(a) Send the engine to an overhaul facility for light overhaul for non-preserved engine.

F. Regular Operation as Alternative to Preservation of Inactive Engines

(1) For engines installed on aircraft, running the engine once a week is an acceptable alternative to avoid the preservation procedures outlined above. This prevents stagnation of fuel in the fuel system, reduces humidity and condensation accumulation in the oil system.

(a) Run engine at 80% TO power or highest attainable for 10 minutes.

G. Accessories, Preservation, Storage and Depreservation

(1) General

(a) Preservation of an accessory depends on the period of inactivity and whether or not the unit is operated during the inactive period.

1 An accessory is considered active if an engine is involved in normal flight operations or if an engine is wet motored or run during preservation periods.

2 If the engine is not wet-motored or operated during the aircraft inactive periods, consider the accessories as inactive.

(b) Accessories that stay on the inactive engines in long-term container storage (off-wing), the time accumulated in the container starts at the time of packaging.

(2) Preservation and Storage:

(a) The preservation procedures that follow are for the accessories removed from the engine and inactive for 28 days or more.

(b) Fill the fuel passages with preservative oil (PWC05-077) for fuel system accessories. Fill the oil passages with engine oil (PWC03-001) for oil system accessories.

NOTE: More procedures are necessary for Honeywell Fuel Control Unit (FCU) (Ref. 73-20-00, FUEL CONTROL UNIT - MAINTENANCE PRACTICES).

1 Drain the unwanted oil, then install the plugs and the covers on all air passages and the fluid passages.

2 Put an applicable shipping cover on all the electrical connections.

CAUTION: DO NOT LUBRICATE THE BLEED VALVE WITH ENGINE OIL (PWC03-001).

(c) Install the covers to seal all openings of the bleed valve.

(d) Put an applicable shipping cover on all the electrical connections of the electrical components.

(e) Put a tag with an applicable accessory information (part numbers, serial number, accumulated time and condition) on the accessory, then put it in a polyethylene bag (PWC05-054). Seal the bag to keep the accessory dry.

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(f) Correctly attach the components in an applicable shipping/storage container with polyurethane foam (PWC05-107). Identify the container with the applicable information from the identification tag.

(g) Store the accessories in a dry location (humidity less than 40%).

(h) Unless specified differently, the storage life limit of accessories is unlimited if preserved as recommended.

NOTE: The storage life limit of the Honeywell Fuel Control Unit (FCU) is six years. If you stored the FCU for more than six years, send the FCU for overhaul.

(3) Depreservation

(a) Remove and retain shipping parts of replacement assemblies for use when returning defective assemblies.

(b) Drain preserving fluids.

NOTE: More procedures are necessary for Honeywell Fuel Control Unit (FCU) (Ref. 73-20-00, FUEL CONTROL UNIT - MAINTENANCE PRACTICES).

(c) Flush replacement assemblies with appropriate system fluid.

7. Storage and Shipping

A. General

CAUTION: TO PREVENT POSSIBLE BEARING DAMAGE SUCH AS BRINELLING, USE OF A "PRATT AND WHITNEY CANADA APPROVED" SHIPPING/STORAGE CONTAINER/SKID WITH SHOCK MOUNTS IS NECESSARY FOR ALL TRANSPORTATION OF AN ENGINE (REF. IPC P/N 3013244, 72-00-00).

The following information and procedures are necessary for storage, preservation and depreservation of an engine . Refer to Removal/Installation for engine installation in the airframe.

NOTE: For engine installation in airframe, refer to REMOVAL/INSTALLATION.

B. Shipping/Storage Container

The shipping/storage container is made up of fiberboard, contains a wooden skid base, a metal cradle is attached to the base, a fiberboard cover, the sides of the container formed with a fiberboard inner and outer sleeve. Plywood sheeting gives more support to the container sides and the cover. You can use the base, plywood supports, cradle and the related hardware again and discard the fiberboard sleeve and cover. Use this container for shipment in a closed conveyance on the North American continent or overseas shipment by air and do not use for shipment by sea. To prevent possible bearing damage such as brinelling, use of a "Pratt and Whitney Canada approved" shipping/storage container/skid with shock mounts is necessary for all transportation (Ref. IPC P/N 3013244, 72-00-00).

The fiberboard container is weather resistant only and should not be exposed to climatic

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conditions for more than seven days and must not be used for outdoor storage. It also offers protection against corrosion for a period of up to six months, provided that the container is correctly sealed and sheltered indoors in relatively dry areas and the color of humidity indicator is monitored every 15 days (Ref. Subpara. D.).

For fiberboard and reinforce fiberboard engine shipping/storage container data, refer to Table 301, Fig. 303 and Fig. 304.

Table 301 Fiberboard and Reinforced Fiberboard Engine Shipping/Storage Container Data

Description	Fiberboard Container Data	Reinforced Fiberboard Container Data
Length	69 in. (1752 mm)	71 in. (1803 mm)
Width	27 in. (686 mm)	35.4 in. (899 mm)
Height	30 in. (762 mm)	37 in. (940 mm)
Weight Empty (approximate)	280 lb. (127 kg)	332 lb. (150.6 kg)

C. Humidity Control

Twelve bags of desiccant material are placed in the protective envelope to maintain internal humidity at a safe level for storage. Each desiccant bag contains eight one-ounce units.

A humidity indicator is installed within the protective envelope and is visible through a window in the fiberboard casing. The moisture level can thus be checked without disturbing the container or its contents.

At a safe humidity level (up to 40 percent relative humidity) the indicator color is blue. As humidity increases the color gradually changes to pink. An all pink color indicates that an excessive moisture condition has been reached; the desiccant must then be replaced with freshly activated bags (Ref. Subpara. D. following).

D. Reactivation of Desiccant and Humidity Indicator

(1) Place bags of desiccant (Ref. IPC) and humidity indicator in a suitable oven controlled at 121°C (250°F). The humidity indicator may be removed when an all blue color has been attained. The desiccant bags should remain in the oven for two hours minimum.

(2) Allow the oven to cool to room temperature (approximately 22°C (72°F)), and then remove the bags and indicator.

(3) Immediately place the desiccant bags in airtight polyethylene envelopes. Exclude all air and heat seal envelopes.

NOTE: Desiccant bags must be removed from the polyethylene envelopes prior to

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installation in the shipping/storage container.

E. Precautions

(1) Before you install or remove engine in/from the shipping/storage container, you must take the necessary precautions as follow:

(a) Make sure a hoist of sufficient lifting capacity, plus safety factor, is available to lift loads as follow:

1 A load of 350 lbs. (159 kg) when you lift the engine only.

2 A load of 800 lbs. (363 kg) when you lift the engine loaded in a fiberboard container.

3 A load of 860 lbs. (400 kg) when you lift the engine loaded in a reinforced fiberboard container.

(b) Put the container or engine correctly on floor directly below the hoist.

(c) Give a free overhead space of 6 feet (2 meters) minimum. Do not consider the distance from the top of the container or assembly to the hook to get 6 feet (2 meters).

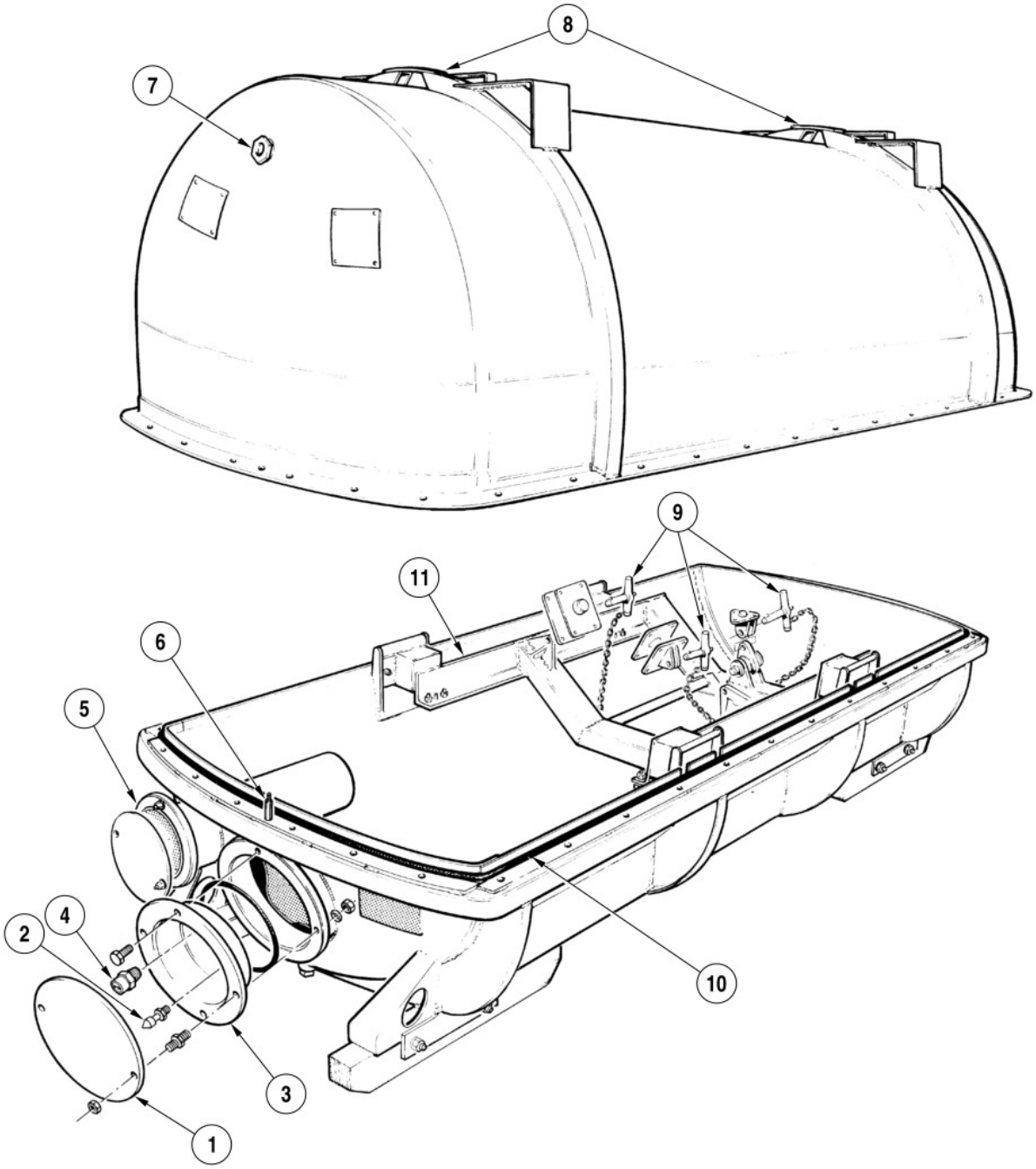
F. Metal Storage and Shipping Container

The metal storage and shipping container is of steel construction, manufactured in two halves and joined at the horizontal centerline with a gasket between upper and lower halves to form a hermetic seal. The container is reusable and may be used for long term indoor and outdoor storage, provided humidity within is frequently checked, desiccant replenished as required and the engine preserved in accordance with the applicable preservation schedule.

The metal container is pressurized through an air fill valve located in the service receptacle, to maintain the pressure within the container at or above ambient pressure. The service receptacle can be removed (Ref. Fig. 302) to provide access to the desiccant cage to facilitate desiccant replenishment. A cylindrical receptacle, located adjacent to the service receptacle, is provided for storage of the engine log book, shipping instructions and other documents.

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Figure 302 Metal Storage and Shipping Container Components



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1. Service Receptacle Cover
2. Air Fill Valve
3. Service Receptacle
4. Pressure Relief Valve
5. Record Receptacle
6. Container Locating Pin
7. Humidity Indicator
8. Container Lifting Points
9. Quick-release Pins
10. Container Gasket
11. Mounting Cradle

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For metal storage and shipping container data, refer to Table 302.

Table 302 Metal Storage and Shipping Container Data

DESCRIPTION	DATA
Length	79 in. (2007 mm)
Width	34 in. (864 mm)
Height	38 in. (965 mm)
Weight	400 lb. (181.4 kg)
Pressure Tested at	10 psig
Material Specification (Skin)	SAE 1010-1020
Paint Primer (One Coat as per)	TT-P-636 or TT-P-664
Paint (Two Coats Enamel as per)	MIL-E-7729

G. Humidity and Pressure Control (Ref. Table 303)

To maintain humidity within the metal container at a safe level for storage, twelve bags of desiccant are tied to the engine, evenly placed over the engine length, 6 on each side. Each desiccant bag contains eight one-ounce units.

A humidity indicator is installed in the service end of the cover, enabling the humidity level to be checked without disturbing the cover. Two viewing windows are provided adjacent to the humidity indicator.

A drain plug is located in the container base just below the service receptacle cover.

At a safe humidity level (up to 40 percent humidity), indicator color is blue. As humidity increases, the color gradually changes to pink. An all pink color indicates that an excessive humidity level exists. Replace the desiccant bags with freshly activated bags (Ref. Subpara. 7. D.). Retain the desiccant bags for reactivation.

Table 303 Temperature/Pressure Gradient

Description	Data							
Ambient Temperature	-28.9°C (-20°F)	-23.3°C (-10°F)	-17.8°C (0°F)	-6.7°C (20°F)	4.4°C (40°F)	15.6°C (60°F)	26.7°C (80°F)	37.8°C (100°F)
Required psi gage pressure	2.000	2.375	2.750	3.500	4.250	5.000	5.750	6.500
NOTE: Five psig is established at a standard working temperature of 15.6°C (60°F).								

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H. Precautions

(1) Before an engine is removed from or installed in a metal container, observe the following precautions:

(a) Before removing an engine from a metal container, loosen relief valve (4, Fig. 302) in the service receptacle to relieve the pressure inside container.

(b) Make sure a hoist of sufficient lifting capacity, plus safety factor, is available to lift loads as follows:

1 A load of 350 lbs. (159 kg) when lifting the engine only.

2 A load of 800 lb. (363 kg) when lifting the engine loaded in a metal container.

(c) Position the container or engine correctly on the floor, directly beneath the hoist.

(d) Allow free overhead space of at least six feet (two meters) exclusive of the distance from top of the container or engine to the hook.

I. Engine Storage in Metal Container (Ref. Fig. 302 and Table 304)

Table 304 Preservation Schedule for Engine in Metal Container

CONTAINER INTERNAL PRESSURE (psi)	**CONTAINER INTERNAL RELATIVE HUMIDITY (%)**	**ACTION REQUIRED**
Normal for prevailing ambient temperature in Table	Below 20 (Indicator all blue)	No action required.
Less than normal for prevailing ambient temperature in Table	Below 20	Repressurize container (Ref. Para. F.). If pressure has dropped after lapse of seven days, remove engine (Ref. Para. E.), and reinstall in a serviceable container (Ref. Para. F.).
	20 to 40 (Indicator partially pink)	Replace desiccant. Repressurize container and check for leakage using a soapy water solution.
	Above 40 (Indicator all pink)	Remover engine from container (Ref. Para. E.). Inspect for corrosion; if condition is satisfactory, reinstall unit with shipping covers, freshly activated desiccant and repressurize container. Check for leakage using soapy water solution. If necessary,

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CONTAINER INTERNAL PRESSURE (psi)	CONTAINER INTERNAL RELATIVE HUMIDITY (%)	ACTION REQUIRED
		rectify faulty container and repeat leakage check. If severe corrosion is present, engine should be shipped to an overhaul facility.

(1) Check the color of humidity indicator at least once every 30 days, dependent upon ambient relative humidity. If humidity indicator color is turning to pink, continue inspection (Refer to Schedule, Table 304).

(2) Remove two nuts, washers and bolts securing the service the receptacle cover (1) to the container base. Remove the cover.

(3) Loosen the air fill valve (2) to depressurize the container.

(4) Remove two lockwashers and nuts securing the service receptacle (3) to the container base. Remove the service receptacle and discard the preformed packing.

(5) Remove eight desiccant bags from the cage using the service receptacle port and retain for reactivation (Ref. Subpara. 7. D.) and reuse. Install eight freshly activated desiccant bags of the same size (8 units per bag) in desiccant cage.

(6) Install a new preformed packing on the service receptacle and insert the receptacle in the port in the container base. Secure the receptacle to the container with two lockwashers and nuts.

(7) Install the service receptacle cover and secure the cover with two bolts, lockwashers and nuts.

(8) Reuse of the humidity indicator should be satisfactory since the color of element is reversible. Replace only if necessary and retain pink indicator for reactivation (Ref. Subpara. 7. D.).

J. Stacking Containers for Storage

(1) Instructions outlined in Table 305 must be followed when stacking empty containers or containers with engines.

Table 305 Stacking Limitations - Engine/Power Section Shipping/Storage Container

METAL CONTAINER		FIBERBOARD CONTAINER	
Engine in Container Three high max.	Make sure that containers are sound and in undamaged condition.	Engine in container Two high Max. (Static Storage) One high Max. (Shipping/Transporting)	Replace used sleeve with new sleeve and cover.
Empty	Only static (no impact)	Empty container	Make sure the wooden side

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METAL CONTAINER		FIBERBOARD CONTAINER	
Container Four high Max.	and superimposed stacking recommended. Avoid dropping containers over each other.	Four high Max.	stiffeners (where fitted) are sound and undamaged. Only static (no impact) and superimposed stacking is recommended. Avoid dropping containers over each other.

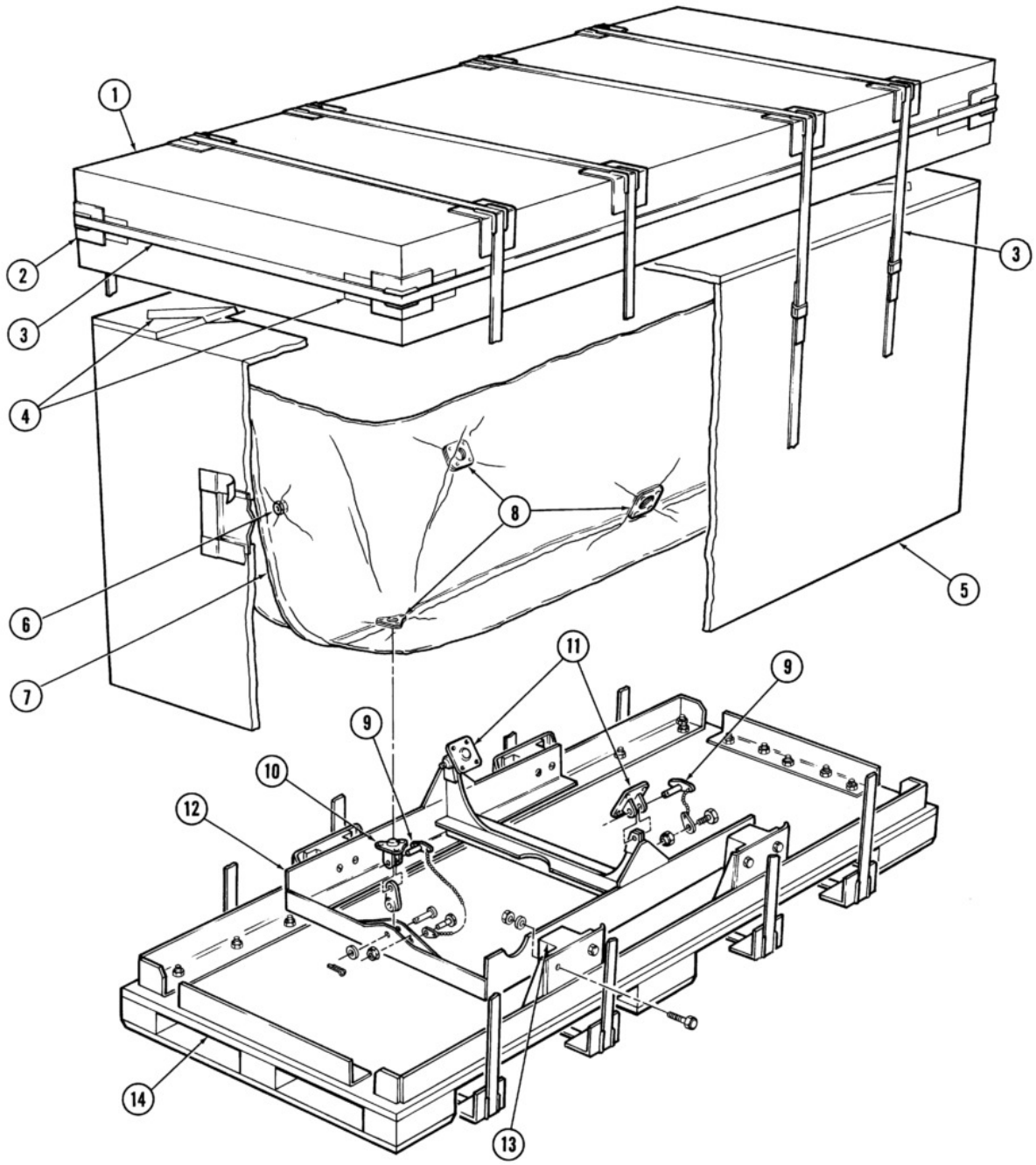
8. Removal/Installation of Engine from/in Fiberboard Container

CAUTION: IF THE ENGINE IS DROPPED, IN THE CONTAINER OR OTHERWISE, RETURN THE ENGINE TO AN OVERHAUL FACILITY FOR LIGHT OVERHAUL. INDICATE "DROPPED ENGINE".

A. Removal

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Figure 303 Engine Shipping/Storage Container



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1. Cover
2. Reinforcing Corners
3. Strap
4. Pressure-sensitive Tape
5. Sleeve
6. Humidity Indicator
7. Protective Envelope
8. Gaskets
9. Quick-release Pin
10. Rear Mount Bracket
11. Front Mount Bracket
12. Cradle
13. Shear Mount
14. Skid Base

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(1) Remove strapping (3) from the fiberboard container. Retain the reinforcing corners (2). Remove the cover (1) and sleeve (5) from the base (14).

(2) Cut the protective envelope (7) and expose the top of the engine.

CAUTION: NEVER USE THE SLING AND LIFTING BRACKETS TO LIFT THE ENGINE ATTACHED TO THE SKID BASE.

(3) Install the sling on hoist; adjust the sling lifting eye for horizontal lift.

(4) Attach the sling to the engine lifting brackets at flanges A and G.

(5) Get the engine weight on the sling. Cut the safety cables then remove the quick-release pins (9) that attach engine to the cradle (12).

(6) Maintain the engine in a horizontal position and raise until clear of the cradle. Remove the engine front mount brackets (11) and rear mount bracket (10), together with the associated hardware, from the engine.

(7) Replace the engine front mount brackets (11) and engine rear mount bracket (10) on the cradle and secure with engine mount pins (9).

NOTE: Retain all other hardware in a bag and secure to the cradle for reuse.

(8) Remove the ignition exciter, if applicable, complete in protective envelope from cradle and retain with the engine.

(9) Remove 12 desiccant bags from the engine and retain for reactivation and reuse (Ref. Subpara. 7. D.).

NOTE: Retain a desiccant bag in the ignition exciter protective envelope until the unit is required for use.When the unit is removed, the desiccant bag should be reactivated for reuse.

(10) Remove the envelopes containing shipping data and the engine log book from the box glued to the skid base.

(11) Remove all shipping caps, plugs, covers and the T5 trim lead tiedown strap from the engine .

(12) Retain moisture barrier/protective wrapping on the engine air inlet screen until the engine is installed in the airframe.

B. Installation (Ref. Fig. 303)

CAUTION: MAKE SURE OIL HAS BEEN COMPLETELY DRAINED FROM THE ENGINE.

(1) With the engine mounted in the stand (Ref. Para. 9.) and the engine preservation procedure implemented (Ref. Para. 6. B.), fit all blanking plates, plugs, caps and the T5 trim lead tiedown strap (Ref. IPC).

(2) Position moisture barrier/protective cover on air-inlet screen with pressure-sensitive tape (PWC05-070).

NOTE: The cover may be made of polyethylene sheet, aluminized cotton-backed material or

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waxed paper.

(3) Make sure the cradle and skid base are serviceable and new or serviceable fiberboard sleeve and cover are available.

(4) Remove the engine from the stand (Ref. Para. A.).

(5) Remove the ignition exciter, if applicable, from the engine and secure the ignition cables with adjustable tie down straps to the engine. Install protective caps to cable ends and secure with pressure-sensitive tape (PWC05-070).

CAUTION: DO NOT USE STANDARD COMMERCIAL HARDWARE TO SECURE ENGINE SUPPORT BRACKETS TO THE ENGINE OR CONTAINER. USE OF HARDWARE OF LOWER STRENGTH THAN SPECIFIED COULD CAUSE SERIOUS ENGINE DAMAGE DURING SHIPPING.

(6) Check that the interior of the protective envelope is free of dirt, oil, water or other contaminants.

(7) Lower the engine into the new protective envelope (7) with gaskets (8) aligned with the lower engine mounting pads each side of the engine and accessory gearbox mounting pad.

(8) Attach the engine front mount brackets (11) to the lower engine mounting pads each side of the gas generator case with eight bolts. Tighten the bolts 225 to 250 lb.in.

(9) Attach the engine rear mount bracket (10) to the mounting pad at the 6 o'clock position on the accessory gearbox with three bolts. Tighten the bolts 50 to 70 lb.in.

(10) Lower the engine into the cradle (12), then attach the front and rear mount brackets to the cradle with quick-release pins (9). Install the safety cables.

(11) Secure the humidity indicator (6) to the protective envelope, so it is visible through the fiberboard sleeve window.

(12) Place the engine log book, log sheets and relevant documents for shipping with engine in a waterproof envelope and seal with pressure-sensitive tape (PWC05-070). Place the envelope in box secured to the skid base and seal the box.

(13) Remove the sling and hoist from the engine.

CAUTION: REMOVE THE DESICCANT BAGS FROM THE PROTECTIVE ENVELOPES BEFORE INSTALLATION.

(14) Tie 12 freshly activated eight-unit desiccant bags (Ref. IPC) to the engine, evenly spaced over the engine length, six on each side.

(15) Heat seal along the top of the envelope using a suitable hand sealing iron, leaving one corner open. Evacuate the air, fold the corner and heat seal.

(16) Put the ignition exciter (Ref. Step (5)) in a protective envelope. Put one four-unit desiccant bag in the envelope. Heat seal the envelope and let one corner open. Attach the envelope to the bottom of the shipping/storage container in the forward left-hand position with a pressure-sensitive tape. Evacuate the air from the envelope, fold the corner and heat seal.

(17) Install the fiberboard sleeve (5) over the engine and fit in position on the skid base (14).

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(18) Fold the engine envelope and make sure that the humidity indicator (6) aligns with the window in the sleeve. Tape the envelope folds with pressure-sensitive tape (4).

(19) Close the flaps of sleeve and seal with pressure-sensitive tape (4).

(20) Place the cover (1) on the sleeve, fold the flaps down and seal at each corner with pressure-sensitive tape (4).

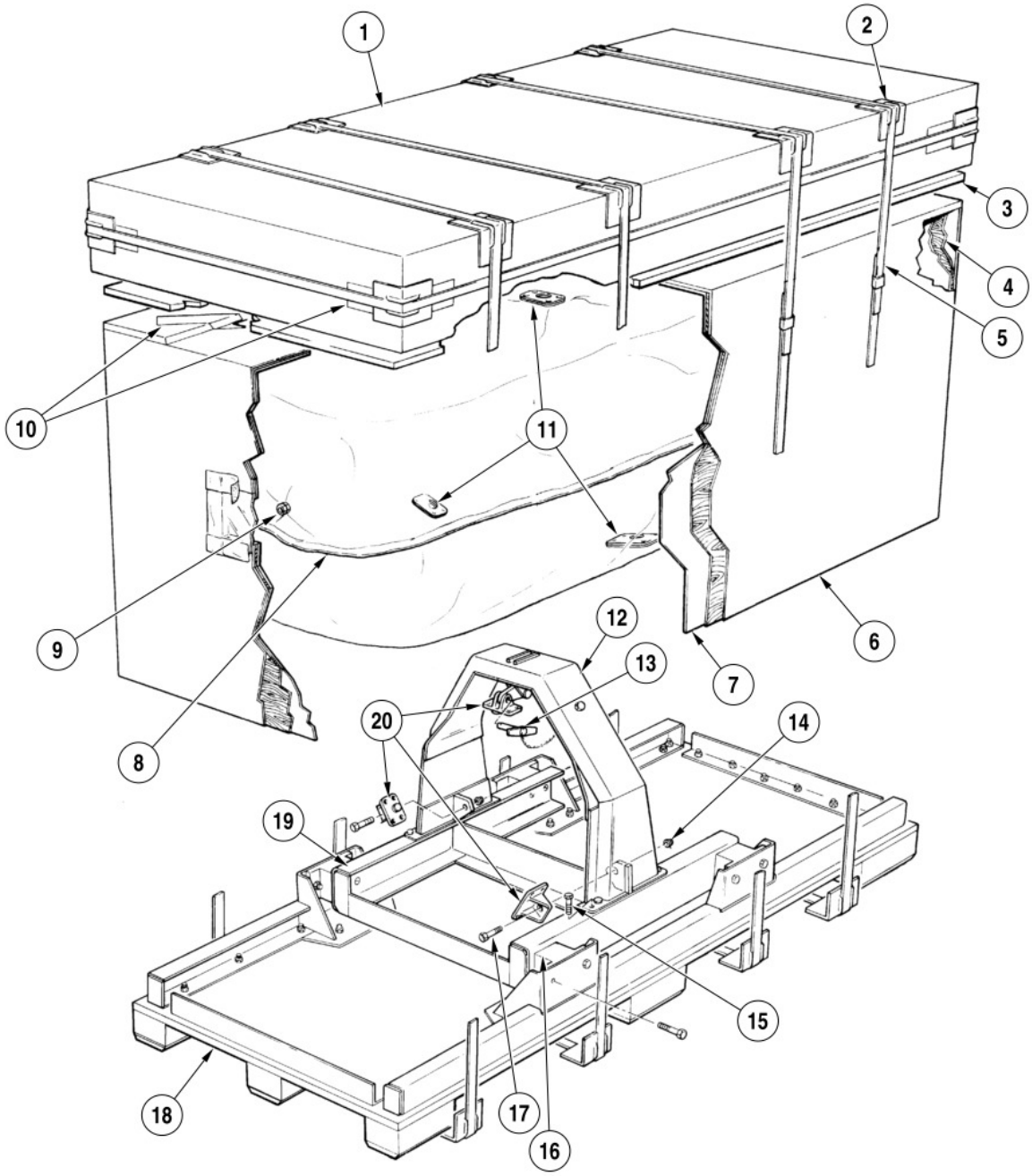
(21) Secure the cover horizontally with strap (3) and four reinforcing corners (2).

(22) Secure the cover and sleeve to the skid base with four straps (3) and 16 reinforcing corners (2).

C. Engine Removal from Reinforced Fiberboard Container (Ref. Fig. 304)

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Figure 304 Reinforced Fiberboard Container Components



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1. Cover
2. Reinforcing Corners
3. Panel
4. Panel
5. Strap
6. Sleeve
7. Sleeve
8. Protective Envelope
9. Humidity Indicator
10. Pressure-sensitive Tape
11. Gaskets
12. Support
13. Quick-release Pin
14. Locknut
15. Bolt
16. Shear Mounts
17. Bolt
18. Skid Base
19. Cradle
20. Mount Brackets

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CAUTION: IF THE ENGINE IS DROPPED, IN THE CONTAINER OR OTHERWISE, RETURN THE ENGINE TO AN OVERHAUL FACILITY FOR LIGHT OVERHAUL. INDICATE "DROPPED ENGINE".

(1) Cut and remove strapping (5) from around the fiberboard container. Retain the reinforcing corners (2). Remove the cover (1) and sleeve (6), panels (4) and sleeve (7) from the base (18).

(2) Cut the protective envelope (8) and expose the top of the engine.

CAUTION: DO NOT USE THE ENGINE SLING AND LIFTING BRACKETS TO LIFT THE ENGINE ATTACHED TO THE SKID BASE.

(3) Install the sling on the hoist; adjust the sling lifting eye for horizontal lift.

(4) Attach the sling to the engine lifting brackets at flanges A and G.

(5) Take the engine weight on the sling; remove the quick-release pin (13), nuts (14) and bolts (17) that secure the support (12) to the engine mounts.

(6) Remove nuts and bolts (15) that secure the support (12) to the cradle (19). Withdraw the support from the engine, taking care not to strike the support against the engine. Remove the engine mounts (20) from the gas generator case.

(7) Maintain engine horizontal and raise until clear of the cradle.

(8) Reinstall the engine mount brackets (20) and engine support (12) on the cradle and secure with the pin (13), bolts (17) and nuts (14).

NOTE: Retain all other hardware in a bag and secure to the cradle.

(9) Remove the ignition exciter, if applicable, complete in protective envelope from the cradle and retain with the engine.

(10) Remove 12 desiccant bags from the engine and retain for reactivation and reuse (Ref. Subpara. 7. D.).

NOTE: Retain a desiccant bag in the ignition exciter protective envelope until the unit is required for use. When the unit is removed, the desiccant bag should be reactivated for reuse.

(11) Remove the envelopes containing shipping data and the engine log book from the box glued to the skid base.

(12) Remove all shipping caps, plugs, covers and the T5 trim lead tiedown strap from the engine .

(13) Retain the moisture barrier/protective wrapping on the engine air-inlet screen until the engine is installed in the airframe.

D. Engine Installation in Reinforced Fiberboard Container (Ref. Fig. 304)

CAUTION: MAKE SURE OIL HAS BEEN COMPLETELY DRAINED FROM THE ENGINE.

(1) With the engine mounted in the stand (Ref. Para. 9.) and engine preservation procedure implemented (Ref. Para. 6. B.), fit all blanking plates, plugs, caps and the T5 trim lead

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tiedown strap (Ref. IPC).

(2) Position the moisture barrier/protective cover on the air inlet screen and install with pressure-sensitive tape (PWC05-070).

NOTE: The cover may be made of polyethylene sheet, aluminized cotton-backed material or waxed paper.

(3) Make sure the cradle (19), plywood panels (3 and 4) and skid base (18) are serviceable and new or serviceable fiberboard sleeves (6 and 7) and cover (1) are available.

(4) Remove the engine from the stand (Ref. Para. A.).

(5) Remove the ignition exciter, if applicable, from the engine and fasten the ignition cables with adjustable tiedown straps. Install protective caps to the cable ends and fasten with pressure-sensitive tape.

CAUTION: DO NOT USE STANDARD COMMERCIAL HARDWARE TO INSTALL THE ENGINE SUPPORT BRACKETS TO THE ENGINE OR CONTAINER. USE OF HARDWARE OF LOWER STRENGTH THAN SPECIFIED COULD CAUSE SERIOUS ENGINE DAMAGE DURING SHIPPING.

(6) Check that the interior of the new protective envelope (8) is free of dirt, oil, water or other contaminants.

(7) Lower the engine into the protective envelope (8) with the gaskets (11) aligned with the engine mounting pads.

(8) Attach the engine mount brackets (20) to the engine mounting pads with bolts. Tighten bolts 225 to 250 lb.in. and install safety cables.

(9) Position the support (12) over the engine, taking care not to strike the engine. Install the support on the engine with quick-release pin (13), bolts (17) and nuts (14). Tighten the nuts 515 to 575 lb.in.

(10) Install a humidity indicator (9) to the protective envelope so it is visible through the fiberboard sleeve window.

(11) Lower the engine and support (12) onto the cradle (19); align flanges of the support with the cradle taking care not to strike the engine against the cradle. Secure the support to the cradle with eight bolts (15) and nuts. Tighten the nuts 190 to 210 lb.in.

(12) Remove the sling and hoist from the engine.

(13) Place the engine log book, log sheets and relevant papers for shipping with the engine in a waterproof envelope and seal with pressure-sensitive tape (PWC05-070). Place the envelope in the box secured to the skid base and seal the box.

CAUTION: REMOVE DESICCANT BAGS FROM THE PROTECTIVE ENVELOPES BEFORE INSTALLATION.

(14) Tie 12 freshly activated eight-unit desiccant bags (Ref. IPC) to the engine, evenly spaced over the engine length, six on each side.

(15) Heat-seal along the top of the envelope using a suitable hand sealing iron, leaving one

corner open. Evacuate the air, fold the corner and heat-seal.

(16) Put the ignition exciter (Ref. Step (5)) in a protective envelope. Put one four-unit desiccant bag in the envelope. Heat seal the envelope and let one corner open. Attach the envelope to the bottom of the shipping/storage container in the forward left-hand position with a pressure-sensitive tape. Evacuate the air from the envelope, fold the corner and the heat seal.

(17) Install the fiberboard sleeves (6) and (7) and plywood panels (4) over the engine and fit in position on the skid base (18).

(18) Fold the engine envelope and make sure that the humidity indicator (9) aligns with the window in the sleeve. Tape the envelope folds with pressure-sensitive tape (10).

(19) Close the flaps of the sleeve and seal with pressure-sensitive tape (10).

(20) Place the plywood panel (3) and cover (1) on the sleeve, fold the flaps down and seal at each corner with pressure-sensitive tape (10).

(21) Secure the cover horizontally with strap (5) and four reinforcing corners.

(22) Secure the cover and sleeve to the skid base with four straps (5) and 16 reinforcing corners (2).

E. Engine Removal from Metal Container (Ref. Fig. 302)

CAUTION: IF THE ENGINE IS DROPPED, IN THE CONTAINER OR OTHERWISE, RETURN THE ENGINE TO AN OVERHAUL FACILITY FOR LIGHT OVERHAUL. INDICATE "DROPPED ENGINE".

(1) Remove two nuts, lockwashers and bolts securing the service receptacle cover (1) to the receptacle and remove the cover.

(2) Loosen the air fill valve (2) to equalize the air pressure before proceeding.

(3) Remove 36 nuts, lockwashers and bolts securing the container top cover to the container base.

(4) Lift the container top vertically to clear the locating pin (6) using the hoist attached to the lifting points (8). Rest the container top on clean wooden blocks.

(5) Attach the sling to the engine lifting brackets at flanges A and G.

CAUTION: DO NOT USE THE ENGINE SLING AND HOIST FITTINGS TO LIFT THE ENGINE SECURED TO THE METAL CONTAINER BASE.

(6) Get the engine weight on the sling. Cut the safety cables then remove the quick-release pins (9) from the engine mounting brackets.

(7) Raise the engine slowly from the container; make sure the engine remains horizontal and is not permitted to swing.

(8) Remove the two engine front mount brackets, rear mount bracket, bolts and washers from the engine mount pads. Retain the brackets and bolts in a suitable cloth or polyethylene envelope and secure to the cradle.

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(9) Remove eight desiccant bags from the desiccant cage and retain for reactivation and reuse.

(10) Remove the self-locking nuts, washers and bolts, and remove the cover from the receptacle (5). Withdraw the engine log book and documents from the receptacle. Replace the cover, bolts, washers and nuts.

(11) Remove all shipping covers, caps, plugs and the T5 trim lead tiedown strap from the engine. Retain the covers, caps, plugs and the T5 trim tiedown strap in a suitable cloth or polyethylene envelope and secure to the cradle .

(12) Clean the interior of the container.

(13) Replace the container top and secure with 36 nuts, lockwashers and bolts.

(14) Coat all exposed hardware with preservative compound (PWC06-012).

(15) Retain the moisture barrier/protective wrapping on the engine air inlet screen until the engine is installed in the airframe.

F. Engine Installation in Metal Container (Ref. Fig. 302)

(1) With the engine mounted in the stand (Ref. Para. 9.) and engine preservation procedure implemented (Ref. Para. 6. B.), fit all shipping covers, plugs and the T5 trim lead tiedown strap.

(2) Locate the moisture barrier/protective cover around the air-inlet screen, making sure there is adequate overlap and full coverage of the screen. Secure the cover with pressure-sensitive tape (PWC05-070).

NOTE: The moisture barrier/protective cover may be made of polyethylene sheet, aluminized cotton-backed material or waxed paper.

(3) Make sure the interior and exterior of the metal container top and bottom halves are clean, dry and free of damage. Make sure a new gasket (10) is available.

CAUTION: MAKE SURE THE OIL HAS BEEN COMPLETELY DRAINED FROM THE ENGINE.

(4) Remove the engine from the stand (Ref. Para. A.).

CAUTION: DO NOT USE STANDARD COMMERCIAL HARDWARE TO SECURE THE ENGINE SUPPORT BRACKETS TO THE ENGINE OR CONTAINER. USE OF HARDWARE OF LOWER STRENGTH THAN SPECIFIED COULD CAUSE SERIOUS DAMAGE DURING SHIPPING.

(5) Remove the two engine front mount brackets from the container and secure the brackets to the engine front lower mount pads on each side of the engine with eight bolts and washers. Tighten the bolts 225 to 250 lb.in.

(6) Remove the engine rear mount bracketfrom the cradle and secure the bracket to the engine rear mount pad with three bolts and washers. Tighten the bolts 50 to 70 lb.in.

(7) Carefully lower the engine onto the cradle (11). Attach the front and rear mount brackets to the cradle with quick-release pins (9) then install safety cables.

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CAUTION: WHEN REACTIVATED DESICCANT BAGS ARE USED, REMOVE THE BAGS FROM THE PROTECTIVE ENVELOPES.

(8) Insert eight bags of desiccant (64 units total) into the desiccant cage. Install the cover and secure. Reuse of the humidity indicator (7) should be satisfactory since the element is reversible. Replace only if necessary.

(9) Liberally coat the new gasket (10) with anti-sticking compound (PWC09-003) and install the gasket in position on the mating flange of lower half of the container.

(10) Lift the top cover of the container into position using the hoist and sling from the lifting points (8). Lower the top cover over the locating pin (6).

(11) Secure the top cover to the container base using 36 bolts, lockwashers and nuts. Tighten the bolts 270 to 300 lb.in.

(12) Coat all exposed hardware with preservative compound (PWC06-012).

(13) Place the engine log book and relevant shipping documents in the receptacle (5). Install the cover and secure with two bolts, washers and self- locking nuts.

(14) Stencil the engine model, serial number, rubber cure date and pertinent preservation and inspection dates on the white panel adjacent to the humidity indicator.

(15) Make sure the service receptacle cover (1) is secure, lockwired and lead sealed.

(16) Introduce clean, dry air through the air fill valve (2) to provide internal pressure equal to value for ambient temperature (Ref. Table 303). Install the valve cap securely.

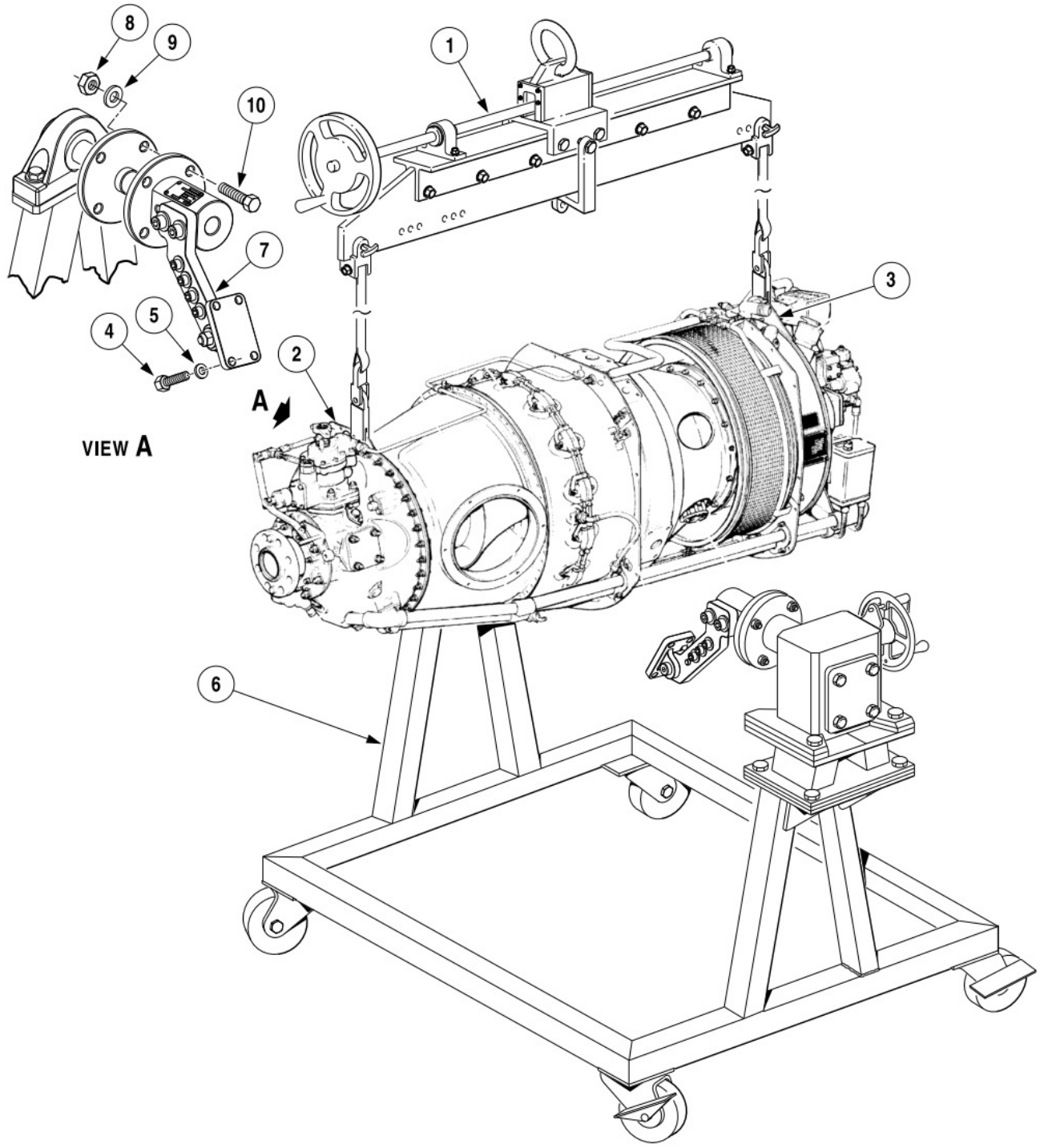
(17) Check the container for pressure drop after twenty minutes. No leaks are permissible. If leaks occur, depressurize if necessary and repeat the sealing procedure until no leaks are detected.

9. Removal/Installation of Engine from/in Maintenance Stand

A. Removal (Ref. Figs. 305 and 306)

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Figure 305 Engine in Stand



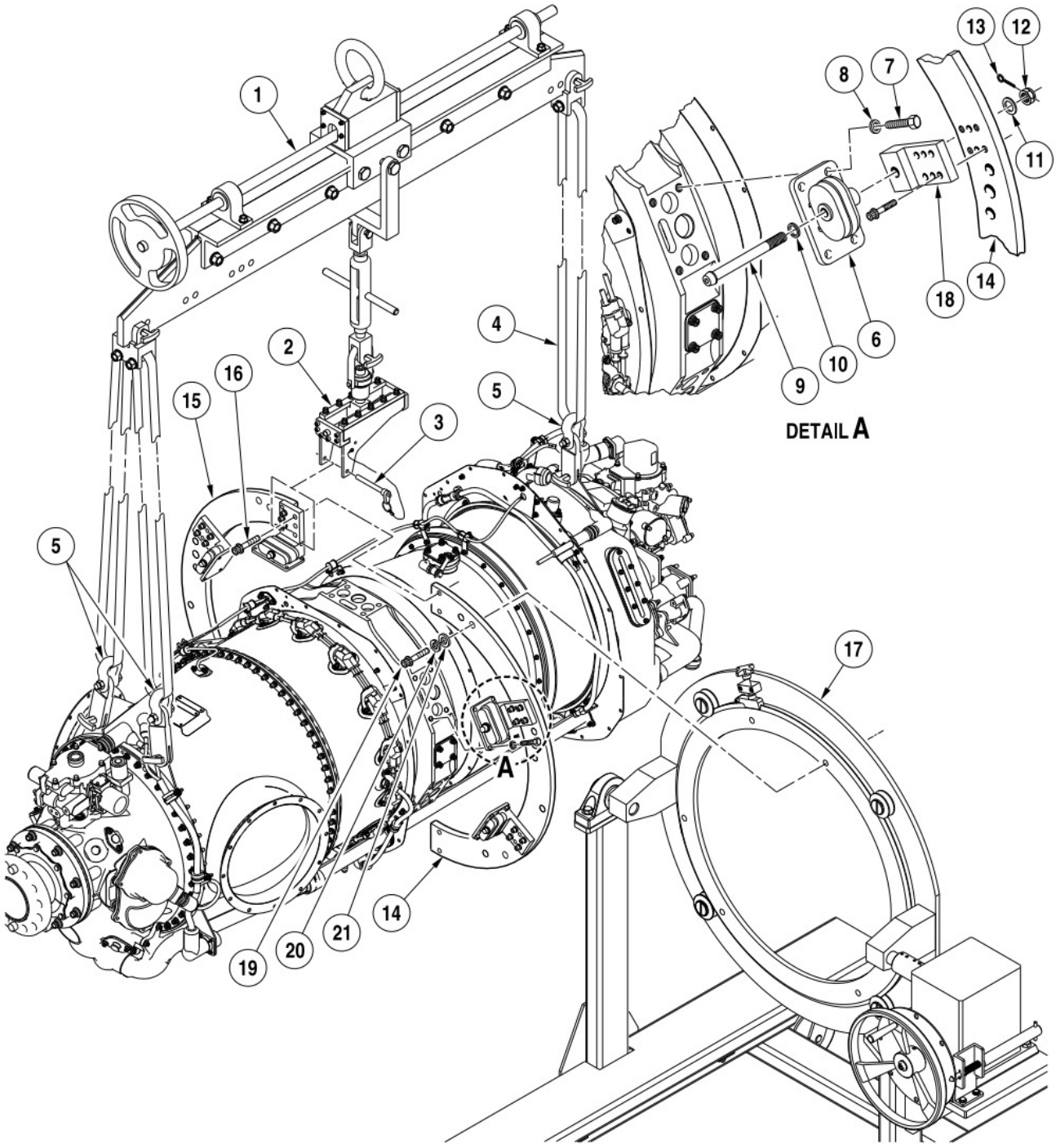
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1. Engine Sling (PWC51861-600)
2. Front Lifting Bracket
3. Rear Lifting Bracket
4. Screw
5. Washer
6. Engine Stand (PWC34300)
7. Adapter (PWC70792)
8. Nut
9. Washer
10. Bolt

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Figure 306 Removal/Installation of Engine from/in Stand



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1. Engine Lifting Sling (PWC51861-600)
2. Engine Lifting Bracket (PWC50373)
3. Release Pin (Part of PWC50373)
4. Shackle Strap (Part of PWC51861-600)
5. Shackles (Part of PWC51861-600)
6. Mount Pads (PWC32080)
7. Bolt (Part of PWC32080)
8. Washer (Part of PWC32080)
9. Bolt (Part of PWC32080)
10. Washer (Part of PWC32080)
11. Washer (Part of PWC32080)
12. Nut (Part of PWC32080)
13. Cotterpin (Part of PWC32080)
14. Mount Ring Half (Left) (PWC70334)
15. Mount Ring Half (Right) (PWC70334)
16. Bolt (Part of PWC70334)
17. Engine Stand (PWC70907 or PWC71835)
18. Lockplate (Part of (PWC70907) or PWC71835)
19. Bolt (Part of (PWC70907) or PWC71835)
20. Lock Washer (Part of (PWC70907) or PWC71835)
21. Washer (Part of (PWC70907) or PWC71835)

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(1) For (PWC34300) engine stand (Ref. Fig. 305):

(a) Attach the sling (PWC51861-600) (1) to the engine lifting brackets (2 and 3) and take up weight of engine with hoist.

NOTE: Adjust lifting eye on sling to maintain engine horizontal during lifting operations.

(b) Remove the screws (4) and washers (5) from adapter (PWC70792) (7) attaching to engine.

(c) Carefully lift the engine clear of the stand (PWC34300) (6).

(d) Put the adapter (PWC70792) (7), screws (4) and washers (5) on the stand (PWC34300) (6).

(2) For (PWC70907) or (PWC71835) engine stand (Ref. Fig. 306):

WARNING: MAKE SURE THAT THE FLOOR LOCK IS FULLY APPLIED IN LOCKED POSITION TO PREVENT MOVEMENT OF THE ENGINE STAND.

(a) Attach the lifting bracket (PWC50373) (2) to the lockplate (18) of the mount ring at 12 o'clock location and attach with the release pin (3).

(b) Attach the sling (PWC51861-600) (1) to the engine lifting bracket (PWC50373) (2) to hold the weight of the engine with the hoist.

(c) Remove the six bolts (19) and the washers (20 and 21) that attach the mount ring halves (PWC70334) (14 and 15) to the engine stand (PWC70907) or (PWC71835) (17).

(d) Remove the engine and lift clear of the stand.

(e) Attach the shackles (5) of the engine lifting sling (1) to the engine lifting brackets at Flanges C and G.

(f) Remove the release pin (3) and remove the engine lifting bracket (PWC50373) (2).

(g) Remove the bolts (16) that attach the mount ring halves (PWC70334) (14 and 15) together.

(h) Remove the cotterpins (13), the nuts (12), the washers (10 and 11) and the bolts (9) that attach mount ring halves (PWC70334) (14 and 15) to the engine mount pads (PWC32080) (6).

(i) Remove the four bolts (7) and the washers (8) that attach engine mount pads (PWC32080) (6) to the gas generator case.

B. Installation (Ref. Figs. 305 and 306)

(1) For (PWC34300) engine stand (Ref. Fig. 305):

(a) Attach engine mount adapter (PWC70792) (7) on the stand (PWC34300) (6) with bolts (10), washers (9) and nuts (8). Tighten the nuts.

(b) Position the stand (6) directly below the engine and carefully lower the engine into the stand. Attach the adapter (PWC70792) (7) on engine with washers (5) and screws (4). Tighten the screws.

(c) Check that the screws are engaged correctly to prevent the engine from tilting.

(d) Remove the sling (PWC51861-600) (1) from the engine.

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(2) For (PWC70907) or (PWC71835) engine stand (Ref. Fig. 306):

(a) With the engine hung in the sling (PWC51861-600) (1), install the four mount pads (PWC32080) (6) at 2, 4, 8 and 10 o'clock positions on the gas generator case. Attach each pad with the bolts (7) and the washers (8). Lubricate the bolts with a dry film lubricant (PWC06-007). Torque the bolts 225 to 300 lbf.in. (25-33 Nm).

(b) Put the mount ring halves (PWC70334) (14 and 15) in position on the gas generator case. Assemble together with the lockplates (18) at 12 and 6 o'clock positions with the bolts (16). Torque the bolts tight.

(c) Attach the assembled mount ring to the correct mount pads (PWC32080) at 2, 4, 8, and 10 o'clock positions with the bolts (9), the washers (10 and 11) and the nut (12). Torque the nut 500 to 600 lbf.in. (56-67 Nm) and attach with cotterpin (13).

(d) Install the engine lifting bracket (PWC50373) (2) on the lockplate (18) of the mount ring at 12 o'clock position and attach with the release pin (3).

(e) Remove the lifting sling shackles (5) from the engine lifting brackets at Flanges C and G for the lifting bracket (PWC50373) (2) to hold the weight of the engine.

WARNING: MAKE SURE THAT THE FLOOR LOCK IS FULLY APPLIED IN LOCKED POSITION TO PREVENT MOVEMENT OF THE ENGINE STAND.

(f) Put the engine into the engine stand (PWC70907) or (PWC71835) (17), attach the mount ring to the stand with the bolts (19) and the washers (20 and 21). Torque the bolts tight.

(g) Remove the release pin (3) and the lifting bracket (PWC50373) (2).

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